

Grade 8
Unit 7
Lesson 5 Practice

Name Answer Key
Date _____
Period _____

The community of Siesta Key, Florida has the fastest shrinking population in the state. In 2010 the population of Siesta Key was 6,568. Then in 2019, the population was 5,570. It was determined that in 2021, the linear pattern continued.

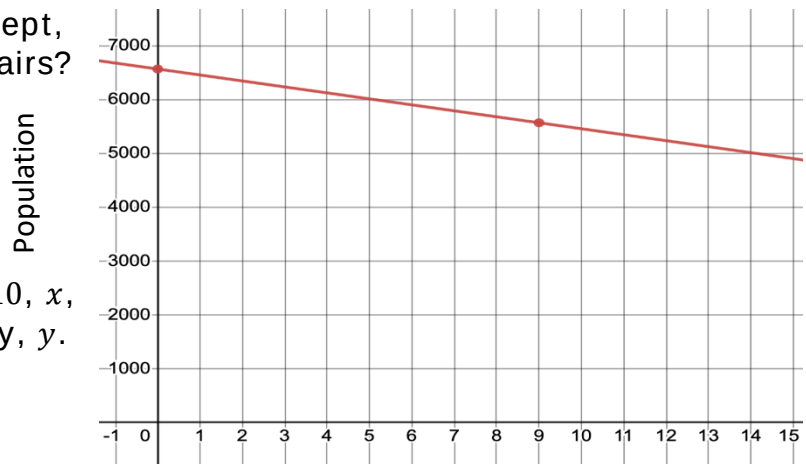
1. Which information about the relationship can be determined from the situation?

- the slope and y -intercept of the graph
- at least two ordered pairs on the graph

2. What are the slope and y -intercept, or at least two of the ordered pairs?

$(0, 6,568)$
 $(9, 5,570)$

3. Graph the linear relationship between time in years since 2010, x , and the population of Siesta Key, y .



The table of values shows how the length of Gabby's book report depends on the number of hours she spends writing.

Hours	Pages
0	0
2	3
6	9

4. What is the y -intercept of the line that is represented in the table of values?

$(0, 0)$

5. What is the slope of the line represented by the table of values?

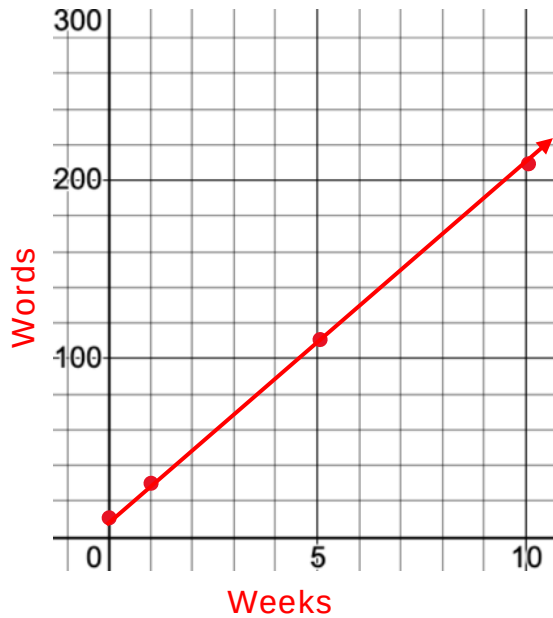
$\frac{3}{2}$ or 1.5

6. Interpret the slope in the context of the situation.

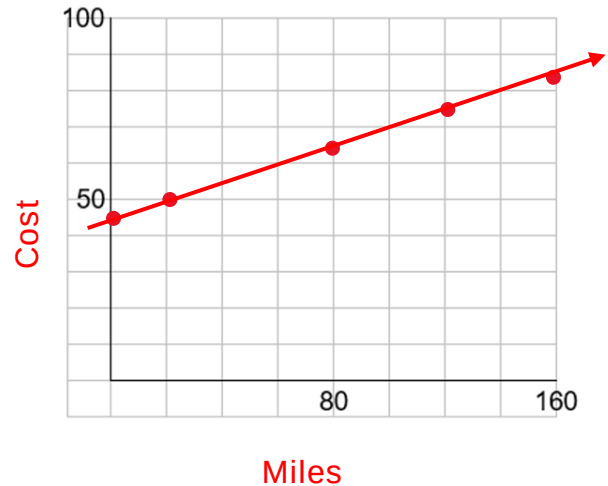
She writes 1.5 pages per hour.

Graph each linear relationship.

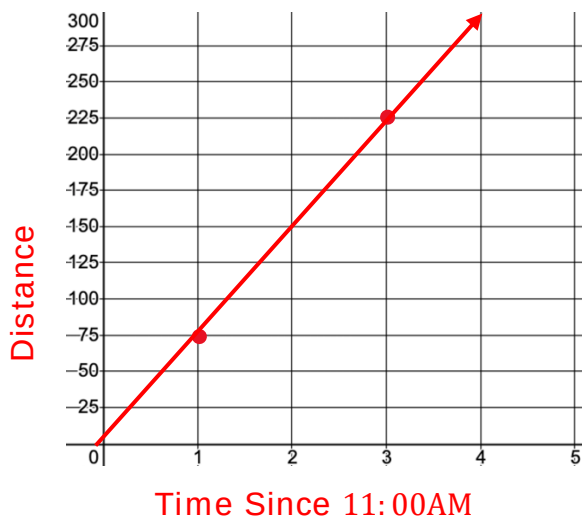
7. Sam is learning how to speak Spanish. He already knows 10 words and plans to learn 20 new words each week. Label the number of words learned, y , and the number of weeks, x .



8. Harlee and her family are on vacation in Mexico City. They are renting a van for a day trip. The base fee to rent the van for the day is \$45. They are then charged \$5 for every 20 miles. Label the total cost, y , and the number of miles, x .



9. We left for our vacation at 11:00 AM at 12:00 AM we had driven 75 miles. Then at 2:00PM our total mileage was 225 miles. Label the miles driven, y , and the time that has passed since 11:00AM, x .



10. Hanan is planning a birthday party. The cost for 8 people is \$105 and the cost for 12 people is \$145. Label the total cost, y , and the number of guests, x .

